

Learn Structure

The lexicon, and the four places my own files use one word twice

“Not every part of a game is strategic. Most of it isn’t.”

website, Learn Structure, opening

Everything in this project follows from that pair of sentences. If most of a long interaction is not strategic, then the expensive machinery is being spent in the wrong places, and the question becomes where the strategic part lives, how to say so in a way that can be measured, and whether the answer can be recovered from data rather than declared by hand.

Three things make this project’s vocabulary harder than the other two in the cluster. The word structure already carries seven precise senses in the literatures it draws on, and I wrote them all down. The house vocabulary changed once, so the corpus carries two names for the same two axes and only one of them is current. And several terms are placed rather than settled, which means an entry has to say what is fixed and stop.

Contents

1	The vocabulary decision, and what it supersedes	2
2	Active	2
2.1	The headword	2
2.2	The two axes	4
2.3	The relevance filter	5
2.4	Structure, in seven senses	7
2.5	The candidate measures	8
2.6	The generator	9
2.7	The learner	11
2.8	The game form	13
3	Working	14
4	Peripheral	15
5	The two checks	16
5.1	Discrimination, run in both directions	16
5.2	Where two of my files disagree	17

6	What crosses into the shared lexicon	17
7	Sources	18

1 The vocabulary decision, and what it supersedes

The AAAI submission fixes the house vocabulary in a drafting-conventions block, and it governs everything below.

“Fixed vocabulary: type (not kind), degree (not amount), dependence”

AAAI27_git/main.tex, drafting conventions

The earlier working file names the two axes the other way in its own section title, and it also uses the third rejected word throughout. Both pairs are therefore live in the corpus and only one is current. An entry that quotes `taxonomy_v2.tex` is quoting superseded wording, and the superseded wording is what a reader will meet first, since that file is the largest in the project at three hundred and eighty-eight kilobytes.

Two consequences, both practical. First, *type* and *degree* are axes, not adjectives, and the classification is type by degree rather than a single scale. Second, the four-level progression from pure independence through state-transition dependence to immediate-reward and future-reward dependence is also superseded. Before the revision the graded axis ran along a single progression, from one-shot play to repeated play to the stochastic game, so it graded one type only, the sequential one. That restriction is gone, and `sections/taxonomy_v2.tex` says so in the paragraph that answers my own question about it.

A dictionary that carries the four-level progression without marking it as retired will send a reader to the wrong axis.

A second naming inconsistency is open and is smaller. The experiments section calls the first domain network defense while the motivating-domains and object-oriented-specification sections call it attack-defense. Both names are in the current draft.

2 Active

Terms I have to define on the spot. All four fields.

2.1 The headword

strategic dependence

“Strategic dependence is how much a player’s best action, in a given region of states and actions, turns on what the other player does.”

website, Learn Structure, the intuition

“Where dependence is strong the region needs game-theoretic reasoning; where it is weak or absent the region collapses into an ordinary single-agent decision.”

website, Learn Structure, the intuition

Definition: The degree to which one player’s best choice depends on another player’s behaviour in a region of states and actions.

Distinguishes it from: The four neighbouring senses of structure, all of which are properties of the game as written: the dependency graph of a factored MDP, the relations of an object-oriented state, the causal graph, and empirical payoff dependence. Strategic dependence is a property of how the game is played.

Operational test: Fix a game completely and vary only the other player's behaviour. If the reasoning the defender requires changes while all four neighbouring quantities stay put, the property in question is this one and none of them can express it.

In my project: The whole project. The separating example is the green security game with two poacher behaviours, one that walks the high-density cells and reads nothing else, and one that conditions its movement on the patrol positions it observes. Against the first, optimal patrolling is prediction and interception. Against the second, the patrol placement changes the behaviour it is trying to intercept.

The same term, defined twice, in payoff units and in best actions

The project page: how much a player's best action turns on what the other player does.

The cluster page, introducing the same project: how much value a defender forfeits by ignoring the opponent, as a function of the game's structure.

Sensitivity of a choice against a loss in payoff units. The two are not the same functional, and the gap has a name inside the project: one of the two recorded holes in the measure set is a pairwise deviation gain in payoff units, and nothing in the candidate set delivers it. So the cluster page is quoting a quantity the project has not yet written down.

One ideal, or a family

The taxonomy's own verdict on the four candidates it keeps: *None of them is strategic dependence; each captures a fragment. The statistical measures detect variation, value-similarity detects consequence for play, influence detects reach, and nothing detects observability.*

And the stronger statement of why the question is not yet askable: whether a candidate is an unbiased estimator of strategic dependence cannot be asked, because strategic dependence has not been written down as a definite functional with its arguments fixed, which game, which policies, which summary. The gap is in the target, not in the candidates.

So the headword is currently a concept with a discriminating test and four partial instruments, and the single-versus-family decision is mine and is not made. An entry that gives a formula here would be inventing one.

pockets

"The dependence is not uniform. It concentrates in pockets, and those pockets move during a single interaction as the opponent shifts."

website, Learn Structure, the intuition

Definition: The regions of the state and action space where strategic dependence is high, which are localized and which move within a single play.

Distinguishes it from: A subgame, whose boundary is a property of the game tree and does not move. A pocket's boundary is a property of the behaviour in force and moves as distances close, observations arrive, and earlier choices commit later ones.

Operational test: Ask whether the boundary could move without the game changing. If it could, the object is a pocket.

In my project: The figure on the project page is a map of exactly this, one panel showing dependence across the space and one probing a single region against four opponent behaviours. The requirement it generates is the first entrance condition on any account: dependence must be a function of state and history, not a number attached to the whole game.

2.2 The two axes

type

“Strategic dependence is classified first by its type: the mechanism through which one player’s best choice comes to depend on another player’s behavior.”

sections/taxonomy.tex

The five, with the page’s own one-line gloss on each:

Type	Carries behaviour through
Time	when a move lands relative to another
Space	where in the state space the two players meet
Control	what one player’s actions make available or deny to the other
Cause	what one player’s actions change that the other depends on
Information	what one player can observe of the other

Definition: The mechanism that carries one player’s behaviour into another’s best action.

Distinguishes it from: Degree, which grades how much dependence a mechanism carries. A degree without a type would say how much reasoning a region needs while withholding which machinery to spend on it.

Operational test: Name the route. If the answer is that today’s play changes which situation is played tomorrow, the type is time. If the answer is proximity, space. If one player’s inputs write variables the other’s outcomes read, control. If one entity’s state enters another’s transition law, cause. If the constraint is what a best response can distinguish, information.

In my project: The five are claimed to cover the shared structure of the three motivating domains, and the scope of the claim is stated with it: no structure of the domains is exhibited that falls outside all five, so the claim holds at the strength of an absent counterexample rather than of a systematic check. A sixth type would enter as one more scale with its own separating tests, leaving the rest in place.

degree

“Each mechanism carries an ordered scale, with a test separating adjacent levels.”

website, Learn Structure, the classification

Definition: The position a structure holds on its type’s ordered scale, whose bottom is the absence of that type of dependence.

Distinguishes it from: A single continuous strength number over all types. The five scales do not share units, and the information scale is a partial order rather than a chain, since two observation structures can each carry something the other does not. There is no faithful aggregate, so a reading is five components, one position per type.

Operational test: An object takes the highest level whose separating test it passes. If no test separates two adjacent levels on a concrete game, the scale is not usable.

In my project: Each scale adopts an external framework where one exists and fills the gap where it does not, and says which is which. Time follows the classical hierarchy of game forms, from one-shot through finitely and infinitely repeated play to the stochastic game. Space runs from the decoupled game through sparse local dependence and structured global dependence to the dense case where no spatial saving remains. Control grades the dimension of the subspace one actor's inputs can steer. Cause follows the layers of the causal hierarchy, with a continuous strength inside the interventional layer given by arrow-cutting relative entropy. Information is ordered by garbling, from the null experiment to full observation.

the zero point

Definition: The bottom of every scale, characterized once: a structure sits at the zero point of a scale exactly when, for that type, every reason a defender could have to attend to it fails.

Distinguishes it from: A stipulated bottom level, which would be a convention of the scale. This one is checked against the four excuses, and four of the five zeros conform outright while the causal zero conforms only with two recorded qualifications.

Operational test: Run the four conditions of the relevance filter at the zero level of the scale. If one of them excuses every structure of that type, the zero point is principled.

In my project: This characterization is the project's own construction and is marked as such, since the adopted frameworks supply the levels of each scale and none of them supplies a shared reading of their bottoms.

2.3 The relevance filter

strategically relevant

The raw form of the idea is mine and is a list of four ways not to care.

“if no variation then no need to care. if variation does not change optimal then no need to care. if out of spatial or temporal range then no need to care. and then, if not observable by the opponent, then no need to care as well.”

sections/plan_dialogue.tex, my own words

The formal positive form on the project page:

“A structure is strategically relevant to a player when four conditions all fail to excuse ignoring it”

website, Learn Structure

Definition: A property of a structure, relative to a player and a game, that holds when all four excuses fail: it varies over its admissible values, its variation changes the player's best response, it lies within reach of play, and the party that would exploit it can observe it.

Distinguishes it from: The strategic relevance of Koller and Milch, which relates two decision nodes of a multi-agent influence diagram and holds when optimizing at one node requires the decision rule at the other. That relation runs between two decision nodes. This one runs between a structure and a player.

Operational test: Every test returns excused or not excused, and no test returns important. A structure needs no game-theoretic reasoning from the player exactly when at least one condition excuses it, and importance is what remains when none does.

In my project: The four are run as a test on a declared triple of the game, the structure under examination, and the structure's admissible value set, with three further parameters declared: the admissible opponent behaviours, the retained conditioning set, and a temporal tolerance. Leaving any of them implicit makes verdicts incomparable across runs.

the four excuses

Definition: No variation, out of reach, opponent-blindness, and invariant best play. Each is sufficient on its own for irrelevance.

Distinguishes it from: A scoring rubric. These are sufficient conditions for dropping a structure, and none of them scores importance.

Operational test: Each has a runnable form. No variation reads the generating process's support over the modelled lifetime, not the snapshot. Out of reach has a spatial clause, structural d-separation from the player's reward and observation variables, and a temporal clause, a discounted tail below a declared tolerance. Opponent-blindness treats the opponent's whole observation process as an experiment and asks whether it is null for the structure. Invariant best play asks whether, for every admissible opponent behaviour, one way of playing in the player's actual strategy set is a best response at every admissible value.

In my project: The logical content is two independent conditions, since a structure is excused exactly when invariant best play holds or opponent-blindness holds. The four survive as a testing order from cheap to expensive: no variation needs only a marginal support, the two reach tests need graph computations and no solving, and invariant best play needs value computation and runs last.

the non-composition caution

Definition: Single-structure verdicts do not compose. A common best play can exist across the admissible values of one structure alone and across those of another alone, and fail to exist across the product of the two sets.

Distinguishes it from: An independence assumption on the structures themselves, which is not what fails. What fails is the transfer of a verdict from one structure at a time to a set dropped together.

Operational test: Ask whether the other retained structures were ranging over their admissible sets while the test ran. If they were held fixed, the excuse is marginal and is revisited whenever structures are dropped together.

In my project: Recorded as the one caution binding the whole list, and it is the reason a filter pass is not a licence to drop everything it excused.

2.4 Structure, in seven senses

structure

My own glossary opens by conceding the problem rather than solving it.

“The word structure carries a different precise meaning in each literature we draw on. The definitions below follow the original sources.”

sections/glossary.tex

Sense	What structure means there
Factored	the sparse dependency graph among state variables across one time step
State abstraction	an equivalence or similarity relation on the state space that lets a decision maker ignore distinctions that do not matter for behaviour or value
Bisimulation	the exact version of that relation, two states equivalent when they have matching reward and matching transition behaviour into corresponding classes
Object	the decomposition of the state into typed objects and their relations
Causal	the directed graph of cause-and-effect relations among variables, actions and rewards, as opposed to merely correlational dependence
Empirical game	the pattern of how each player’s payoff depends on the other players’ strategies inside the estimated game
Strategic (in)dependence	the degree to which one player’s best choice depends on the other player’s behaviour, in a given region of states and actions

Definition: A set together with a relation. The dependence is the relation the structure carries, and the structure is the set plus that relation.

Distinguishes it from: Dependence alone, which is a relation without the set it is carried on. The distinction is what lets a structure be named as a thing to test: a state variable, a payoff term, an edge of the interaction graph, a signal channel, a mechanism parameter.

Operational test: Ask which of the seven the sentence needs, and whether the answer changes if the players change their behaviour. Only the seventh does.

In my project: The seven fall into three families of my own grouping. Factored, object and causal structure are statements about the environment. State abstraction and bisimulation are statements about equivalence. Empirical game structure and strategic dependence are statements about the game. Or, in one line: how the world factors, which distinctions to keep, and who must reason about whom. And the explicit non-commitment, which the dictionary carries too: no claim is made that any one sense reduces to another.

2.5 The candidate measures

the candidate set

Five lineages, seeded from game abstraction because that is the body of work that built smaller games by exploiting non-uniform dependence.

Candidate		My own one-line reading
Graphical games		Drops links between players. Keeps no edge between two players whose payoffs do not affect each other. The graph is fixed by the modeler.
Influence-based abstraction	ab-	Drops outside state variables, keeps only what crosses into the local region. The kept set is chosen by a graph test.
Value-similarity abstraction	ab-	Merges states whose future-strength trajectories are similar.
Information-theoretic influence		Marks the state and action pairs where one agent changes the other's transition or reward. Zero means independent there.
Attention in multi-agent RL	multi-	Drops links between agents, then weights the survivors. The cut is learned end to end by a gate.

Definition: The existing quantitative purchase on strategic dependence, assembled as the set the classification has to account for.

Distinguishes it from: A set of competing definitions of one thing. These are objects of three kinds: a map of where interaction exists, a ruler inside one player's view, and three meters that output amounts.

Operational test: Ask what each takes in and what it puts out. A game alone; a game plus fixed outside policies plus a region; a game plus a joint policy; one training run.

In my project: The comparison is organized by property rather than by candidate, and the sharpest finding comes out of the opponent property: none of the four is defined for an adapting adversary. The graphical representation presupposes no behaviour at all, the influence construction requires the other players' policies fixed and known, the information-theoretic pair presupposes teammates with a shared reward, and attention presupposes co-trained agents inside one run. Against the three domains, where the intruder, the evader and the poacher all adapt, every candidate needs the adversarial adaptation.

Four candidates and a contrast case, or five candidates

The status file says five, and lists value-similarity abstraction among them.

The taxonomy section says four, and excludes value-similarity as the contrast case, since it measures the quality of an abstraction, which is a within-agent question and not dependence between players.

The exclusion argument is the better one and it is the one on the project page. The count is still live in the corpus in both forms, and the decision on whether value-similarity stays inside the set or sits outside it as the contrast case is mine and is not made.

attention, left unplaced

“Attention is left unplaced, because no result ties its score to a quantity of the game.”

website, Learn Structure, where the measures sit

Definition: The one candidate with no anchored placement on the scales.

Distinguishes it from: A measure placed at the zero point of a scale. Expressing a type at degree zero and not reading the type at all are different facts, and assigning attention a zero would assert the first when only the second is known.

Operational test: Ask what a zero weight means. If nothing in the literature says whether it means absence of dependence, the score has no referent and cannot be placed.

In my project: The status changed during the project, from candidate to instrument. The blocker on adapting it to a moving opponent is structural rather than technical, since there is no defined target quantity to adapt.

the two holes

“Nothing measures agent-to-component causal strength by edge-cutting, and nothing measures regret per ordered pair of players.”

website, Learn Structure, where the measures sit

Definition: The two recorded gaps in the measure set, given as what a new member would have to deliver.

Distinguishes it from: A completeness claim. Nothing in the document shows that no further member is missing, and the two are what is recorded rather than what is exhaustive.

Operational test: For the first, feed a target an independent copy of a source and read the effect. For the second, compute regret for one ordered pair of players. Exploitability is the nearest existing quantity and does not serve, because it is global and defined over whole strategies rather than per pair.

In my project: The causal scale grades exactly the first quantity and no member of the set delivers it, which is the clearest case in the project of a scale outrunning its instruments.

2.6 The generator

set by construction, measured

“Every structure the generator reports falls in exactly one of two cases.”

sections/generator.tex

Definition: Two, and only two, grounds on which a reported structure’s value is known. Set by construction: written directly as a generator parameter, true because it was set. Measured: it emerged from what was built, and its value is estimated afterward on the generated instance, which serves as its own referent, with sample counts reported beside it.

Distinguishes it from: A third case, which does not exist. A record entry whose truth rests on neither construction nor measurement would be an assumption wearing the clothes of a fact, and

the whole evaluation chain above it would inherit the assumption.

Operational test: Ask where the number came from. A parameter, or a probe. If neither, the request is refused.

In my project: The measured case is not a concession. Part of strategic dependence is a property of play rather than of the game as written, since the same game yields one realized dependence under one pair of policies and none under another. The rules set a ceiling and play decides how much of the ceiling appears, which is why the generator has a strategy stage at all and why its final stage measures rather than sets.

the decoupled base

Definition: A base instance in which each entity's transition rule reads only that entity's own attribute and its own driver's action component, so the base provably contains no cross-entity dependence, with every dependence introduced afterwards as one explicit rule edit.

Distinguishes it from: The alternative route, which generates a richly connected instance and verifies the requested structure by testing. That route fails on the expensive direction, because soundness requires showing not only that the requested dependences hold but that no unrequested dependence crept in, and establishing absence by testing means testing every pair at every lag.

Operational test: Ask whether auditing the edits audits the instance. It does exactly when the inserted edge is the only path by which the dependence can appear.

In my project: One recorded exception. The adversary's reward reads the joint state, as harvest value at the poacher positions, because a fully decoupled reward would be constant and would fail the validity screen requiring rewards to vary under play.

refusal

Definition: The generator's response to a request it cannot express or realize, given with its notion of infeasibility named, rather than clipped to the nearest attainable value.

Distinguishes it from: Silent approximation, which would put a value in the record that nobody requested and nothing guarantees, and would then grade the learner against a truth that is itself an approximation of unknown quality.

Operational test: Five notions, and a refusal names one. Not specifiable, when the request falls outside the form's vocabulary. Not attainable, when the request is well formed but no instance of the declared shape realizes it. Not orderable, when the requested degree does not sit on the type's ordered scale. Not verifiable, when the achieved degree could not be checked by any probe within a declared budget. And not knowable in a real domain, when the request's referent is a quantity no real counterpart could supply.

In my project: One approximation is licensed and only one. Requests about played dependence name a target for a quantity the generator does not set, so the pools are grown toward the target, the realized map is measured, and the miss is published as a number rather than absorbed.

the perturbation operators

Definition: The vocabulary of ways a handed structure can be wrong, used to build the perturbed conditions of the experiments. Six: declaring a present dependence absent, declaring an absent dependence present at a sampled level, reversing direction on a directed type, relabelling the

type with the degree mapped to the same relative position, shifting the degree without crossing the bottom, and carrying a stale record across an instance change.

Distinguishes it from: Adding noise to the record. Each operator maps a record to a record the edit forms could have produced on a neighbouring instance, so the learner is handed a possible truth rather than a malformed object.

Operational test: Ask whether the perturbed record is well formed under the same grammar the true record uses. If it is not, the perturbation is not one of these six.

In my project: The degree shift is deliberately barred from crossing the bottom, so that strength errors never look like absence errors. A seventh exists in the vocabulary without being run, entity mis-individuation, which is the one error the given entity decomposition makes invisible.

2.7 The learner

the manipulated setting

“One learner, two manipulated parties.”

sections/learn_stage.tex

Definition: The defender is the sole learner. The attacker and the benign party are manipulated: their policies are set by the experimenter, fixed within an episode batch, and varied across batches on a known schedule. They do not learn.

Distinguishes it from: Both sides learning, where the target moves and nothing single-agent learning theory says about stationary environments applies.

Operational test: Ask whether the transition law the learner faces is stationary within a batch. Under manipulation it is, because the other two policies fold into the environment.

In my project: The price is stated openly. Every learned quantity is relative to the manipulated policies in force when the data was collected, which makes it a property of that configuration rather than of the game. The compensation is that the manipulation schedule is the experiment’s intervention channel on the opponent side: the defender’s own actions are interventions and the attacker’s are not, so a quantity defined interventionally over attacker variables becomes estimable only through the schedule.

fidelity

Definition: Divergence between environment and model. Roll model and environment from the same starts for a given number of steps and measure the divergence of the resulting state distributions, reported as a curve in the number of steps rather than as one number.

Distinguishes it from: The training objective, which is whatever the optimizer descends and is a property of the solution rather than of the problem. The training objective is never the reported measure of success.

Operational test: Ask at what horizon the number was measured. A fidelity number without its horizon is not readable, since small per-step divergence still permits value error growing with the square of the effective horizon.

In my project: Chosen in total variation because that is what the model-based control bound

consumes, and reported with the step count for the same reason.

readback

Definition: Comparison of a recovered structure against the planted parameter of the generated game, one error measure per target quantity.

Distinguishes it from: Fidelity, which compares rollouts. The two answer different failures: a model can minimize its training objective and fail fidelity, having fit the data and not the law, and it can pass fidelity while failing readback, its rollouts matching while its internal structure mislabels who drives whom. Fidelity constrains only the joint law, not its factorization.

Operational test: Ask whether the check reads the model’s insides or only its outputs. Only readback reads the insides, and it is available only where a planted parameter exists.

In my project: Structural Hamming distance for the edge relation, defined on equivalence-class representatives so that differences no data could distinguish are not penalized. Sup-norm error for the value metric. An error in the first norm for the influence channel, which doubles as an ingredient of the value-loss bound. Absolute error for the two information-theoretic measures.

controllable latent, exogenous latent, opponent embedding

Definition: The three-way split of each device’s dynamics by driver. The controllable latent is the part driven by the defender’s actions. The exogenous latent is the part driven by background processes no agent controls, user activity, service fluctuations, routine noise. The opponent embedding is inferred from a history of tokens at that device.

Distinguishes it from: IFactor’s four-way split, which cuts by two axes at once, controllable against not and reward-relevant against not, and proves each block identifiable. The split used here is the coarser two-way one, and whether to adopt the reward-relevance axis is not settled.

Operational test: Cut the action signal out of one pathway and ask whether anything in the architecture changes. Gradient routing is what keeps the exogenous pathway isolated from the action signal.

In my project: The policy operates on the controllable and opponent components only, so it does not react to uncontrollable environmental variation, while the reward decoder still conditions on the exogenous latent, so imagined rewards reflect its contribution.

the two attention graphs

Definition: Two parallel self-attention operations across device slots. The controllable one has each device attend to others’ controllable latents and to its own opponent embedding, and its weights carry strategic dependence, conditioned on where the attacker is. The exogenous one has each device attend to others’ exogenous latents with no opponent access, and its weights carry environmental dependence from shared background processes.

Distinguishes it from: A single interaction graph, which is what every alternative found keeps. Keeping two is the distinguishing choice, and the reason is that environmental correlation must not be read as strategic dependence.

Operational test: Find a device pair scoring high on the second graph and low on the first. That pair is environmentally linked and not strategically linked, and it can still be treated as a separate subgame.

In my project: Both weights change within an episode, which gives a time-resolved picture of which pairs become strategically dependent as the attacker moves and which stay only environmentally correlated. That is the learned counterpart of the pockets on the project page.

exogenous

The term hunt is recorded, and the wrong word is named.

“Noise is indeed wrong: noise is uncorrelated with reward, whereas this source is exogenous but reward-coupled, which is the whole difficulty.”

`cards/GenerateMDP-03-terminology-a-exogenousname.tex`

Definition: A state variable or process the agent’s actions do not control, which affects the agent through the reward and through the endogenous state transitions.

Distinguishes it from: Noise, which is uncorrelated with reward. The hard case here is reward-correlated, and calling it noise loses the difficulty that makes the factoring worth doing.

Operational test: Ask whether the process moves the reward. If it does and the agent cannot steer it, the word is exogenous and not noise.

In my project: The specific hard case has its own name, a reward-correlated distractor, for the overlap with the reward-relevant or opponent-affected bits. The generator does not yet accept declared exogenous processes as a generation input, which is a gap because one planned experiment hands the defender part of the environment’s own dynamics while the strategic remainder stays unknown, and that condition needs the environment’s own dynamics to be a separately recorded object.

2.8 The game form

the five game features

Definition: What the three motivating domains share: sequential play, compositional or graph-structured state, adversarial objectives, heterogeneous players, and endogenous environment dynamics.

Distinguishes it from: A list of desirable properties. Each feature imposes one entrance condition on any account of strategic dependence, under one warrant written once: a feature enlarges what can happen in play, and an account blind to the enlargement returns one verdict where the games separate into many.

Operational test: For each feature, name the pair of plays it makes possible and ask whether the account separates them.

In my project: The benign party is part of the fourth feature and is not decoration. It is always present in the real domain, users, civilians, bystanders, and prior two-party formulations omit it, which is a modelling limitation rather than an absence from the games.

three-party, heterogeneous players

Definition: The naming for the setting, deliberately chosen over multiagent. Three-party places the work in the literature. Heterogeneous players names the structural property, players asymmetric in objective and observation.

Distinguishes it from: Multiagent, which is too generic and says nothing about asymmetry, and two-party, which omits the benign side.

Operational test: Ask whether the third party has an objective of its own and a different observation channel. If it does, the setting is three-party and heterogeneous.

In my project: The caveat travels with the term. The naming comes from the cyber and Stackelberg literature, and pursuit-evasion and green security games may name the third party differently or lack one.

compositional

Definition: Two senses, and both are live. In the state sense, a single game’s state divides into parts, a network of devices or a grid of locations, which is a property of the state. In the game sense, a large game splits into smaller games, which is what compositional means in game theory.

Distinguishes it from: Factored, which is the state sense with the parts fixed as variables. The ambiguity here is not between factored and object representations. It is between a property of the state and a property of the game.

Operational test: Ask what is being divided, the state or the game.

In my project: The project uses the state sense in its feature list and inherits the game sense from the literature it cites, which is exactly the kind of place a reader will substitute one for the other without noticing.

MOO-MDP

Definition: The multi-agent object-oriented Markov decision process, chosen as the representation over a factored MDP.

Distinguishes it from: A factored MDP, which fixes a variable count. The object-oriented form gives per-host entities that recur across states, supports a varying number of objects, and shares one class-level transition function per class, which gives transfer across states and across object instances. In the multi-agent form the transition depends on the joint action and each agent carries a private reward.

Operational test: Ask whether the state has a fixed number of slots. If adding a host adds a variable rather than an instance of an existing class, the representation is factored rather than object-oriented.

In my project: The reasons are all matching reasons: entities match the compositional feature, varying object count matches the domain, class-level sharing matches the transfer the learner wants, and private rewards match the heterogeneous-player generation.

3 Working

Recognized precisely, lookup acceptable.

empirical game-theoretic analysis

“in empirical game-theoretic analysis every payoff entry is estimated by simulation, already prohibitive beyond two players, so which strategies enter the empirical game governs what the

analysis can deliver”

website, Learn Structure, why it forces a choice

Distinguishes it from: Solving a game given its payoff function. Here the payoff function is the expensive object and is estimated, so the analysis is bounded by the simulation budget before it is bounded by the solver.

restriction error, sampling error

Definition: The two error sources of a restricted strategy set, of which only one is bounded. Sampling error is estimated payoffs against true payoffs on a fixed strategy set, and it is bounded, since an equilibrium of the estimated game is an approximate equilibrium of the true game over the same set. Restriction error is the chosen subset against the full policy space, and it is not bounded.

Distinguishes it from: One another, and the distinction is the whole point. Restricting the strategy space is standard rather than cheating, and generation is simpler under a restricted set because we plant it, while validation is harder because a richness argument then has to be supplied.

green security game

Definition: The domain of the worked counterexample and of the implemented generator: a grid park whose cells carry an animal density, poacher units driven by the adversary, and patrol units driven by the defender.

Distinguishes it from: The network defense domain, which shares the five features and differs in whether proximity is read on a terrain or on a graph.

qcard

Definition: One node in a tree of questions and partial solutions, with a stable identity assigned once and never reused, a parent, an order, and three metadata fields kept distinct: nature is the cognitive move, polya is the classical heuristic, and method is the named conventional procedure.

Distinguishes it from: A note. A card never holds a fabricated answer next to a real one, and where the answer is not known the field holds the next action or the child identifiers.

proof-generated, convenience-generated

Definition: Two ways a definition comes to exist, following Lakatos. A proof-generated concept is a definition whose content is exactly what some theorem needs, repaired in response to counterexamples. A convenience-generated one is chosen by a designer for a purpose no theorem records.

Distinguishes it from: A judgement of quality. The distinction is predictive rather than evaluative: the two proof-generated objects in the candidate set are the two with theorems, and that is the whole explanation of the guarantee pattern.

4 Peripheral

Roughly where it belongs, retrieved when needed.

HetOvlp

A placeholder in the maturity table for the case of multiple heterogeneous agents affecting overlapping bits of one object. The term is open and I have not found the established one.

action-based, strategy-based

Placed as a property of a profile rather than as a level on any of the five scales, and recorded per candidate. The placement is written down and my own note says I am still not sure where it fits, so the entry stops at the placement.

the five holes

What the taxonomy leaves deliberately open: a per-agent-pair interventional strength for adversarial play, a graded directional functional dependence on strategy spaces, a pairwise deviation gain in payoff units, an identification analysis for every surviving quantity under one-sided observation, and the layer that turns dependence realized in play into bounds on the game. Each is stated with the components its filler must assemble and the theorems it must prove.

the attack set

From Kiekintveld's line of work, and the verdict is that it is not a sixth type. It is a region selector, a rule marking which parts of the game carry equilibrium weight and which can be dropped. That role is the one the strategic-dependence map plays, and the difference is what each marks by: the attack set marks by equilibrium value, the map marks by opponent-dependence.

strategy clustering

Also not a new type. It is an equivalence relation on strategies defined by payoff invariance, and what is new relative to the current shape of the work is the object it operates on, since the project's structures are stated over states or variables and this one is stated over strategies.

5 The two checks

5.1 Discrimination, run in both directions

If one entry names a second as its neighbour, the second names the first.

Entry	Its neighbour
strategic dependence	the four senses of structure as written
type	degree
degree	the zero point
strategically relevant	the four excuses
the four excuses	the non-composition caution
set by construction	measured
the decoupled base	refusal
fidelity	readback
controllable latent	exogenous latent
the two attention graphs	pockets
exogenous	noise
compositional, state sense	compositional, game sense
restriction error	sampling error

Two distinctions run one way only and both survive the check rather than failing it. Strategically relevant names Koller and Milch’s strategic relevance as its outside neighbour, and that relation names nothing back, because it was fixed for a different pair of objects and has no reason to know about this one. Attention names a placed measure as its neighbour and no placed measure names attention, because the asymmetry is the finding: one side of that pair has a referent and the other does not.

5.2 Where two of my files disagree

The disagreement	The entry it sits in
Best-action sensitivity or forfeited value	strategic dependence
One ideal or a family of measures	strategic dependence
Four candidates and a contrast case, or five	the candidate set
Type and degree, or the superseded pair	the vocabulary decision
The four-level progression, current or retired	degree
Network defense or attack-defense	the vocabulary decision
State sense or game sense	compositional

Four of the seven are one problem seen from four sides: the project changed its vocabulary once and the largest file in it was not rewritten. That is a housekeeping matter and it is also the reason a reader meets the retired words first.

6 What crosses into the shared lexicon

Terms this project shares with the rest of Cyber World Modeling. Their cluster sense is in `cwm_shared.tex` and the entries above carry only what Learn Structure fixes about them: world

model, factoring, latent, opponent latent, non-stationary, drift, identifiability, strategic dependence, regret, and empirical game.

Two of those differ across projects and stay in both files. Regret here is the per-ordered-pair quantity nothing in the measure set delivers, while in AcceleratePSRO it is the regret of a profile against a strategy set, and those are different functionals with one name. Drift here is a scheduled change made by the experimenter and named the manipulation schedule, while in FOE-Dreamer it is an unscheduled change made by an adversary and is the thing a detector is supposed to catch. Same movement, opposite epistemic position.

7 Sources

- The website page `strategic-structure.md` and the Cyber World Modeling overview, for the intuition, the classification table, the four conditions, the placement of the measures, and the two holes.
- In the project folder, `Notes/Latex/sections/`: `glossary.tex` for the seven senses and the three families; `strategic_dependence.tex` for the separating example, the five entrance conditions and the Koller distinction; `taxonomy.tex` for the five types, the degree scales, the zero point, the four irrelevance conditions and the candidate comparison by property; `taxonomy_v2.tex` for the correction to the graded axis and for the target-quantity gap; `generator.tex` and `generator_algorithm.tex` for construction against measurement, the decoupled base, refusal and the perturbation interface; `learn_stage.tex` and `learner.tex` for the manipulated setting, fidelity, readback and the three identifiability limits; `current_status.tex` for the maturity table, the five lineages and the two attention graphs; `experiments.tex` for the perturbation operators and the recorded scope of the one experiment that has runs.
- The cards under `Notes/Latex/cards/`, for the one-line gloss on each of the five lineages, the relevance filter in its runnable form, the proof-generated and convenience-generated split, the restriction and sampling error pair, and the two verdicts on Kiekintveld's structures.
- `Notes/Latex/sections/plan_dialogue.tex`, for my own raw statements of the four excuses and of the graded axis, quoted as they stand.
- `Submissions/[July 28th] AAAI (attempt)/AAAI27_git/main.tex`, for the fixed vocabulary and for the recorded ending-condition failure.
- The four fields, the three tiers and the rule on terms that cross projects are from `_side/_lexicons/lexicon_meth`